Ratnajeet Patil

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LinkedIn | LeetCode | GitHub | Portfolio

EDUCATION

DES PUNE UNIVERSITY — June 2024 - June 2027
Computer Science BTech — Pune, Maharashtra

SOU VENUTAI CHAVAN POLYTECHNIC — September 2021 - June 2024
Computer Technology Diploma — Pune, Maharashtra

EXPERIENCE

IEEE COMPUTER SOCIETY DES PU STUDENT BRANCH | Vice Chair (Founding Member) — Pune Maharashtra | Nov 2025 – Present

- Co-founded the branch; organizing open-source workshops and tech community events for 100+ students

IEEE DES PU STUDENT BRANCH | Web Master — Pune Maharashtra | Feb 2025 – Present

- Built and maintain the branch website using React.js; coordinating web presence for IEEE Pune events

SUJALAM CHEMICALS | Software Developer — Pune, Maharashtra | June 2023 – August 2023

- Built a full-stack cross-platform inventory management app using React Native (TypeScript), Express.js, and SQLite, deployed on both Android and iOS
- Designed responsive UI components and RESTful API endpoints, reducing manual stock tracking effort for 3 staff members
- Implemented real-time data sync using Ngrok tunneling for local backend access across devices

SKILLS

Programming Languages	Python, Java, JavaScript, TypeScript, C++, C, CSS
Libraries/Frameworks	React.js (TypeScript), React Native, Flask, Express.js
Tools / Platforms	Git, Github, VS code, Linux
Databases	SQL, MYSql, MongoDB, SQLite

PROJECTS / OPEN-SOURCE

DIQA: NO-REFERENCE IMAGE QUALITY ASSESSMENT LIBRARY | Link — Python

- Published open-source Python library for automated image quality scoring without reference images
- Implemented XGBoost-based routing that dynamically selects the most accurate scoring method per image
- Built a custom training pipeline (DIQATrainer) so users can train models on their own datasets
- Integrated with pyiqa for feature extraction; extensible to additional IQA models

MODULAR LLM SCORER — Python, OpenAI API, Gemini API, Groq API, OpenRouter, Ollama,SciPy

- Built a modular framework to benchmark LLMs across Math, Logic, Knowledge, and Code domains.
- Implemented statistical evaluation using 95% Wilson Confidence Intervals and McNemar's Test with secure sandboxed code execution.
- Supported 8 LLM providers including OpenAI, Gemini, Groq, OpenRouter, and Ollama.

AUTOMANIM | Link — Python, Flask, Manim, Groq API, HTML/CSS, JavaScript, FFmpeg

- Built a full-stack web application that converts natural language into executable Manim animation code
- Integrated Groq LLM API to parse user descriptions and generate Python animation scripts automatically
- Implemented Flask backend for real-time code execution and video rendering in under 10 seconds
- Designed responsive frontend with live preview of rendered mathematical animations

ORDER AND INVENTORY MANAGEMENT | Link

JavaScript, React.js, React Native,CSS

This is an app for managing the chemical inventory of Sujalam Chemicals. It was developed as a project for an internship. The app is built using React Native and Expo Router.

Features

- Keep track of your chemical inventory in real-time
- Easily add, edit, and remove chemicals from your inventory
- Generate reports and analytics to help you make informed decisions
- Dashboard for quick access to important information
- Inventory management for tracking and organizing your chemicals
- Order management for handling orders and shipments

VISITOR ENTER AND APPROVAL APPLICATION | Link

Android Studio, Node.js, Express.js

An Android-based application for efficient visitor registration and approval management.

Key Features:

- Visitor Registration: Captures details including name, contact info, ID proof, and photo.
- Approval Workflow: Enables authorized users to approve or reject visitor requests.
- Notifications: Sends real-time status updates to visitors.
- Data Storage: Secures visitor information in a Room database.
- User Authentication: Provides secure login for personnel.

SMART ATTENDANCE MANAGEMENT SYSTEM | Link

Arduino UNO,ESP8266,RFID module, LCD Display 16x2,I2c

- Built an IoT attendance system using RFID card scanning, LCD display, and Wi-Fi data sync via ESP8266
- Automated attendance logging for 50+ students, eliminating manual register entry
- Synced scan data to a cloud sheet in real-time over Wi-Fi using ArduinoJson and ESP8266WiFi libraries

CERTIFICATIONS

- Python - Kaggle
- Java - HackerRank
- JavaScript - HackerRank
- CSS - HackerRank
- SQL - HackerRank
- Node - HackerRank
- Problem Solving - HackerRank
- Intro to Machine Learning - Kaggle
- Cybersecurity - LinkedIn

HONORS & AWARDS

- Presented research paper "Adaptive Hybrid Architecture for NR-IQA" at ETFI Conference (pending IEEE Xplore publication)
- Authored Modular LLM Scorer: A Rule-Based and Statistical Pipeline for LLM Benchmarking (manuscript in preparation; conference submissions ongoing).
- Published a paper on RFID based Attendance System
- Finalist of IEEE R10 Innovation Challage
- Web Developer/Volunteer for IEEE Pune Congress